Quantum Drug Discovery for Malaria & Tuberculosis Using VQE + Quantum ML

Executive Summary:

A hybrid quantum–classical pipeline accelerating drug discovery using VQE, QSVM, and QGAN.

Objectives:

1. Use VQE for molecular energy calculations.

2. Apply QML for drug-likeness prediction.

3. Generate molecules using QGAN.

4. Provide scalable African biomedical pipeline.

Methods:

- Qiskit Nature for Hamiltonians

- VQE with UCCSD ansatz

- QSVM/QNN for classification

- QGAN for molecule generation

Impact:

Accelerates drug discovery from years to months and supports Africa ' s health sector.